



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

USB TO ETHERCAT ADAPTER BOARD

The domain of the Project:

PCB DESIGN

COURSE NAME:

PCB DESIGN

Team Mentors (and their designation):

Mr. Paramesh Kumar

Team Members:

Ms. Ishwari Sampatakumar Bilagi

Period of the project

March 2026 to September 2026

Declaration : I hereby declare that the project titled “USB to EtherCAT Adapter Board” has been completed by me as part of the PCB Design course under the guidance of PARAMESH KUMAR, Team Mentor. I confirm that this project work was carried out by me during the project period March 2026 to September 2026 as part of my practical learning in PCB Design.



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

The project is titled “USB TO ETHERCAT ADAPTER BOARD .” has been mentored by Paramesh Kumar organised by SURE Trust, from April 2023 to August 2023, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Ms. I.S.Bilagi.

Mentor’s Name : Mr.Paramesh Kumar
Designation—Wakefit

Prof. Radhakumari
Executive Director & Founder
SURE Trust



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

Table of contents

- 1.Executive summary
- 2.Introduction
- 3.Project Objectives
- 4.Methodology & Results
- 5.Social / Industry relevance of the project
- 6.Learning & Reflection
- Future Scope & Conclusion



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

Executive Summary

The **USB to EtherCAT Adapter Board** is a PCB design project developed as part of the **PCB Design** course under SURE Trust – IERY. The main objective of the project is to design a PCB that supports the required hardware interface between USB and EtherCAT communication.

The project was developed using **KiCad**, covering the complete PCB design process from schematic development to PCB layout. The work included component selection and placement, schematic design, PCB routing, layer configuration, silkscreen arrangement, 3D visualization, Design Rule Check (DRC), Gerber generation, and Bill of Materials (BOM) generation.

The final PCB design was verified using KiCad's **Design Rule Checker**, with **zero DRC errors** obtained. The required manufacturing files, project files, PCB images, 3D views, and BOM were also prepared as part of the project deliverables.

This project provided practical experience in PCB design and helped develop an understanding of the complete workflow involved in converting a circuit concept into a manufacturable PCB design.



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

Introduction

2.1 Background

Printed Circuit Board (PCB) design is an important part of electronic product development. A PCB provides the physical platform for connecting electronic components and establishing the required electrical connections between them.

The **USB to EtherCAT Adapter Board** project focuses on developing a PCB design for an interface between USB and EtherCAT-related hardware. The project involves the complete PCB design workflow, starting from schematic development and continuing through PCB layout, routing, verification, and preparation of manufacturing files.

2.2 Problem Statement

The project aims to develop a properly designed and organized PCB for the USB to EtherCAT Adapter Board. The design needs to ensure that the required components, connections, PCB layers, routing, and physical layout are correctly implemented while satisfying PCB design rules.

2.3 Scope of the Project

The scope of this project includes:

- Designing the circuit schematic.
- Creating the PCB layout using KiCad.
- Selecting and placing the required components.
- Routing the PCB tracks.
- Configuring PCB layers and design rules.



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

- Adding required silkscreen information.
- Checking the PCB using Design Rule Check (DRC).
- Generating Gerber files for PCB manufacturing.
- Generating the Bill of Materials (BOM).
- Creating 2D and 3D views of the PCB for documentation.

2.4 Limitations

This project mainly focuses on the **PCB design and documentation** of the USB to EtherCAT Adapter Board. The report covers the design and verification activities performed in KiCad. Physical PCB manufacturing and hardware testing are outside the scope of the design work.

2.5 Innovation Component

The project provides practical experience in developing a complete PCB design workflow for a USB to EtherCAT Adapter Board. It combines schematic design, PCB layout, routing, verification, 3D visualization, and manufacturing-file preparation into a single design process.

The template specifically asks the Introduction to cover the **background/context, problem statement or goals, scope and limitations, and innovation component**.



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

Project Objectives

3.1 Main Objectives

The main objectives of the **USB to EtherCAT Adapter Board** project are:

- To design the schematic of the USB to EtherCAT Adapter Board using KiCad.
- To develop the PCB layout based on the designed schematic.
- To place and route the electronic components properly on the PCB.
- To configure the required PCB layers and ensure proper copper routing.
- To perform PCB design verification using KiCad Design Rule Check (DRC).
- To generate the required manufacturing files, including Gerber and drill files.
- To generate the Bill of Materials (BOM) containing the components used in the design.
- To create 2D and 3D views of the PCB for design documentation and visualization.
- To gain practical hands-on experience with the complete PCB design workflow using KiCad.

3.2 Expected Outcomes

The expected outcomes of the project are:

- Complete KiCad project files
- Schematic design & PCB layout
- Verified PCB design
- Gerber and drill files Bill of Materials
- PCB layout images & 3D PCB views



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

Methodology and Results

4.1 Methodology

The USB to EtherCAT Adapter Board was developed by following a systematic PCB design workflow. The process started with schematic development, followed by component placement, PCB layout design, routing, design verification, and generation of manufacturing and documentation files.

The PCB design was developed using **KiCad**. After completing the schematic and PCB layout, the design was checked using the **Design Rule Checker (DRC)**. The required Gerber files, drill files, Bill of Materials (BOM), PCB layout images, and 3D views were then generated for documentation and manufacturing purposes.

4.2 Tools and Software Used

KiCad

KiCad was used as the primary PCB design software for the project.

The following KiCad features were used:

- **Schematic Editor** — for creating the circuit schematic.
- **PCB Editor** — for designing the PCB layout and routing.
- **3D Viewer** — for viewing the PCB in 3D.
- **Design Rule Checker (DRC)** — for checking PCB design errors.



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

- **Gerber/Drill File Generation** — for preparing manufacturing files.
 - **BOM Generation** — for generating the component list.
-

4.3 Project Workflow

The project followed the workflow below:

Schematic Design



Component Placement



PCB Layout



Routing



Design Rule Check (DRC)



3D Verification



Gerber & Drill File Generation



BOM Generation



Final Documentation



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

4.4 Project Architecture

The USB to EtherCAT Adapter Board consists of several functional sections designed to work together as part of the overall PCB. The major sections of the design include the **STM32 microcontroller, LAN9252 EtherCAT controller, USB interface, RJ45 interfaces, I²C EEPROM, and power supply section.**

4.4.1 STM32 Microcontroller

The STM32 microcontroller forms one of the main processing sections of the design. The schematic includes the microcontroller along with its associated clock, reset, power, and signal connections.

4.4.2 LAN9252 EtherCAT Controller

The LAN9252 forms the main EtherCAT controller section of the design. It is connected to the required power, clock, control, and communication signals.

4.4.3 RJ45 Interface

The design contains two RJ45 connector sections for the Ethernet/EtherCAT interface. These provide the physical network connections required for EtherCAT communication.



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

4.4.4 USB Interface

A dedicated USB interface is included in the design to provide the USB connection and required USB signal connections.

4.4.5 I²C EEPROM

An I²C EEPROM section is included for non-volatile memory and is connected through the required I²C signals.

4.4.6 Power Supply

The design includes a power supply section with a **5 V to 3.3 V conversion stage** and supporting capacitors for the circuit.

4.4.7 Overall Architecture

The major functional flow of the design can be represented as:

USB Interface → STM32 Microcontroller → LAN9252 EtherCAT Controller → RJ45 EtherCAT Interfaces

The **power supply section** provides the required supply voltages to the circuit, while the **I²C EEPROM** provides non-volatile memory.



Figure 1: Complete Schematic of the USB to EtherCAT Adapter Board

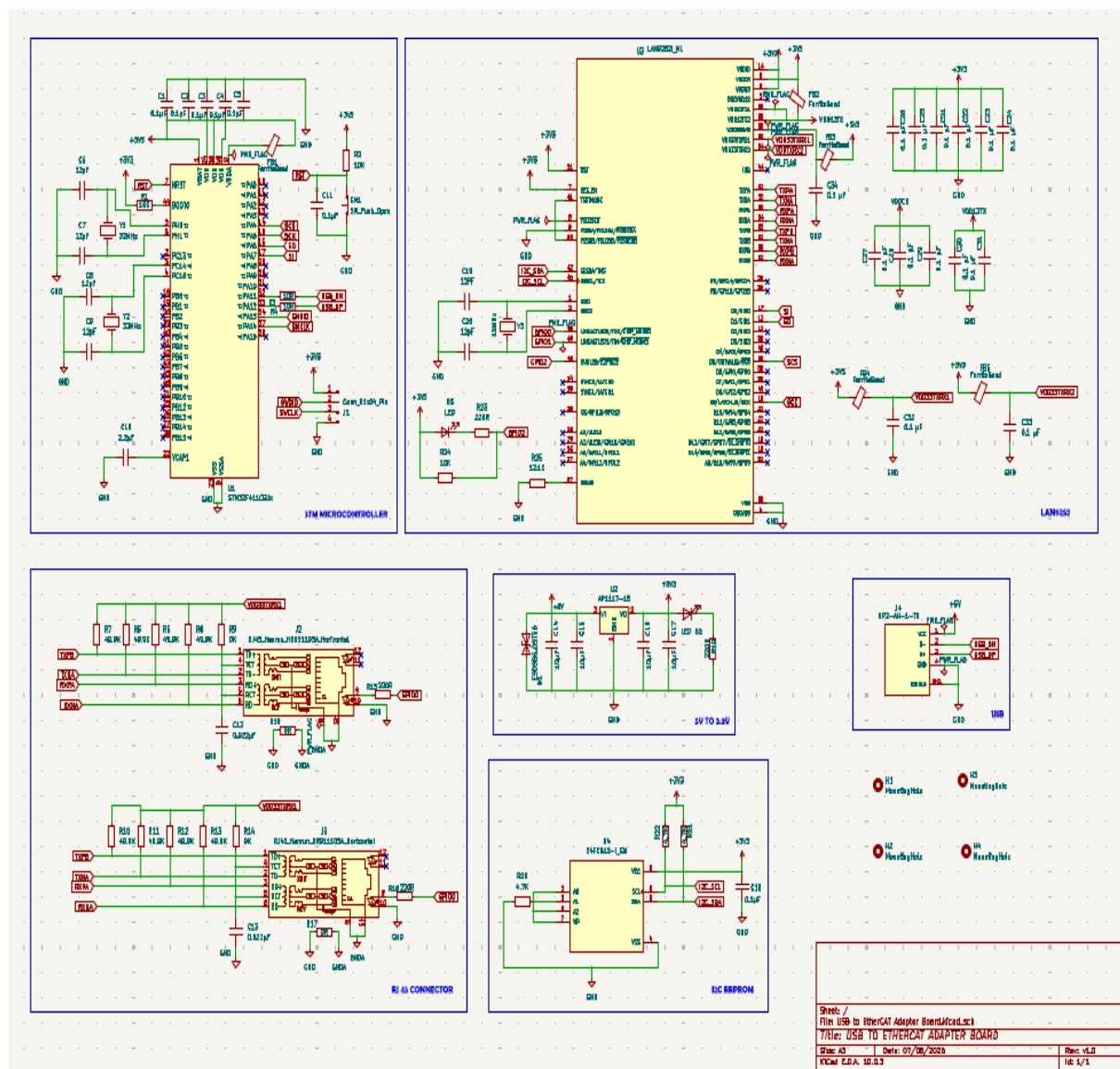


Figure 1 shows the complete schematic of the USB to EtherCAT Adapter Board, including the STM32 microcontroller, LAN9252 EtherCAT controller, USB interface, RJ45 interfaces, I²C EEPROM, and power supply sections.



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

4.5 PCB Layout Design

Below the Figure 1 explanation, type:

4.5 PCB Layout Design

The PCB layout was developed using the KiCad PCB Editor based on the completed schematic. The components were placed according to their functional requirements and the available board area.

The PCB layout includes the required component footprints, copper tracks, vias, mounting holes, and board outline. Routing was performed using the front and back copper layers to establish the required electrical connections.

The completed PCB layout was then checked using KiCad's Design Rule Checker before generating the manufacturing files.



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

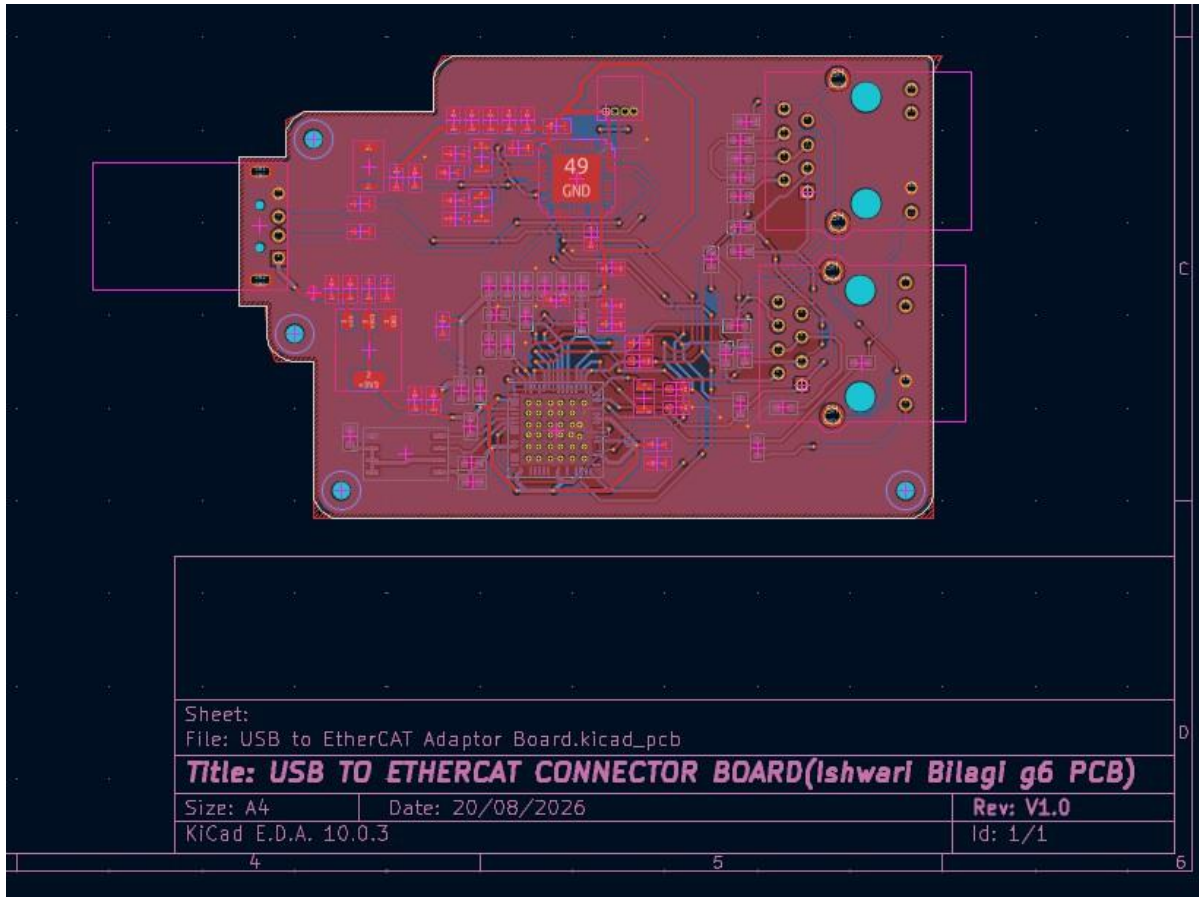


Figure 2 shows the completed PCB layout of the USB to EtherCAT Adapter Board in KiCad, including the placed components, routed copper tracks, vias, mounting holes, and defined board outline.

4.6 Design Rule Check (DRC)

Below Figure 2, type:

4.6 Design Rule Check (DRC)



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

After completing the PCB layout and routing, the design was verified using the **Design Rule Checker (DRC)** in KiCad. The DRC was used to identify possible design-rule violations in the PCB layout.

The final DRC verification resulted in **zero errors**. Some warnings were present, including a warning related to silkscreen being clipped by the board edge.

The zero-error result indicates that no DRC errors were reported in the final PCB design.

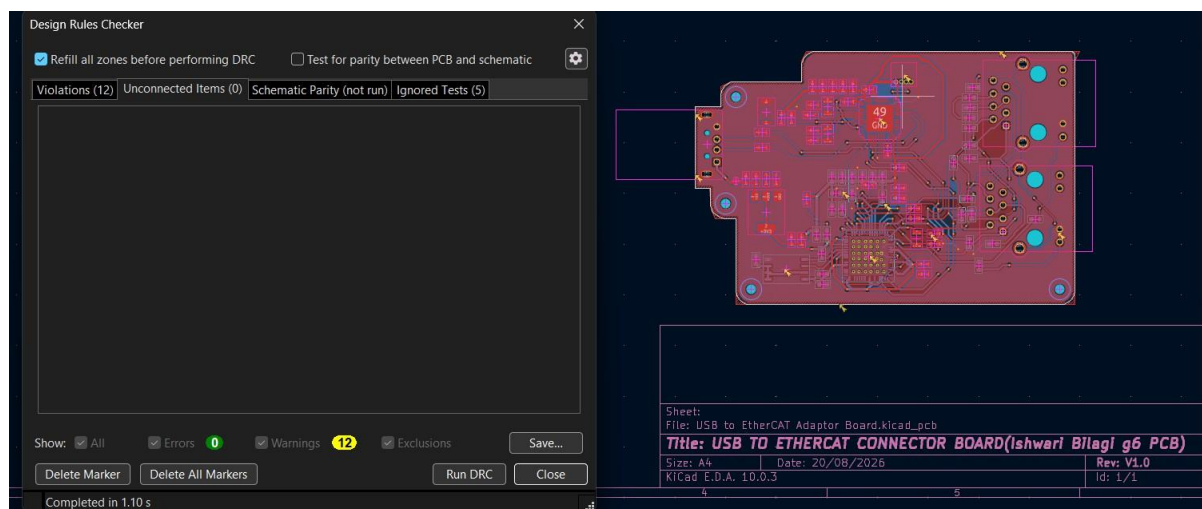


Figure 3 shows the final Design Rule Check result of the PCB design. The design contains zero DRC errors, while the reported warning relates to silkscreen clipping at the board edge.



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

Figure 4: +3V3 Power Routing in the PCB Layout

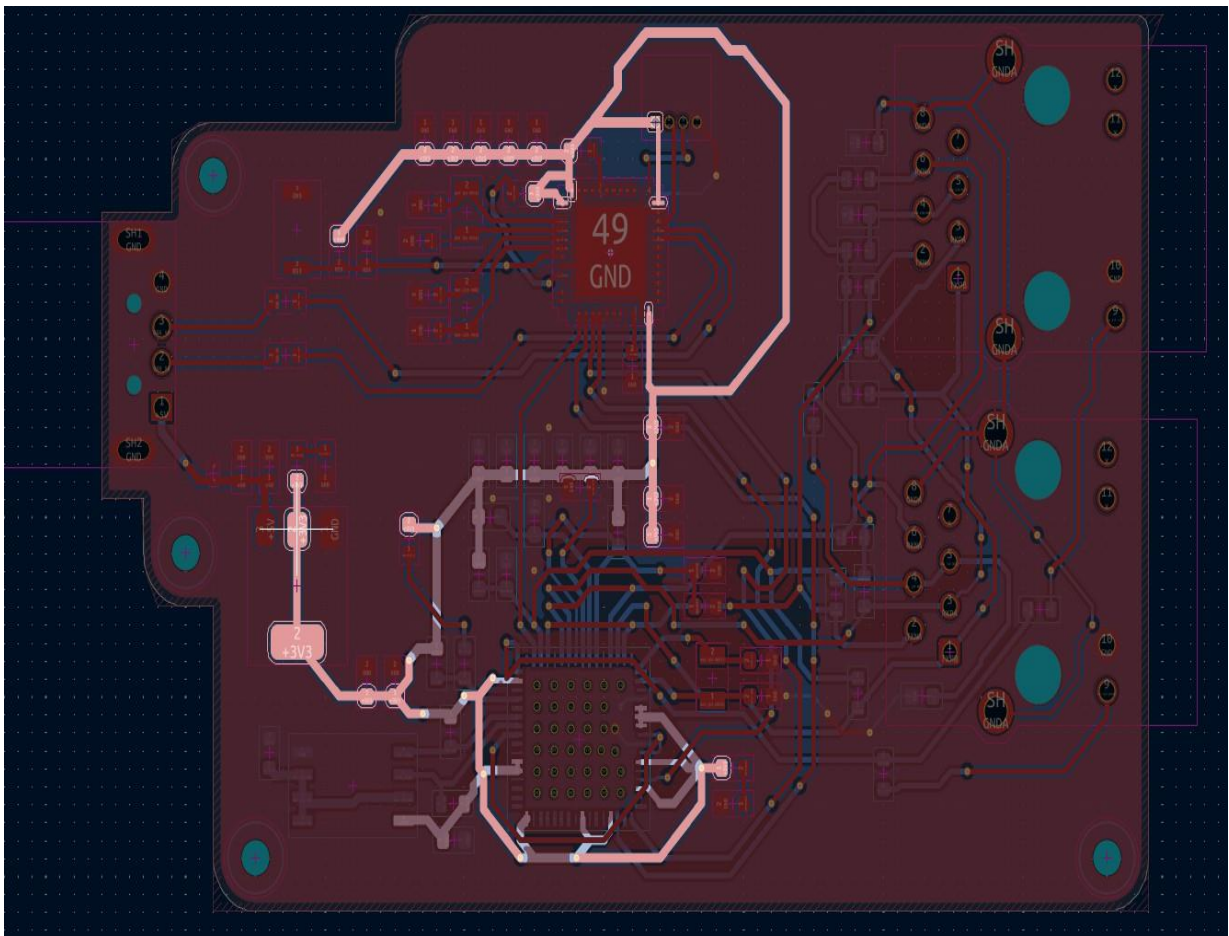


Figure 4 shows the +3V3 power routing in the PCB layout. The highlighted traces show the distribution of the 3.3 V supply across the board to the required circuit sections.



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

Figure 5: Top Layer (F.Cu) of the PCB

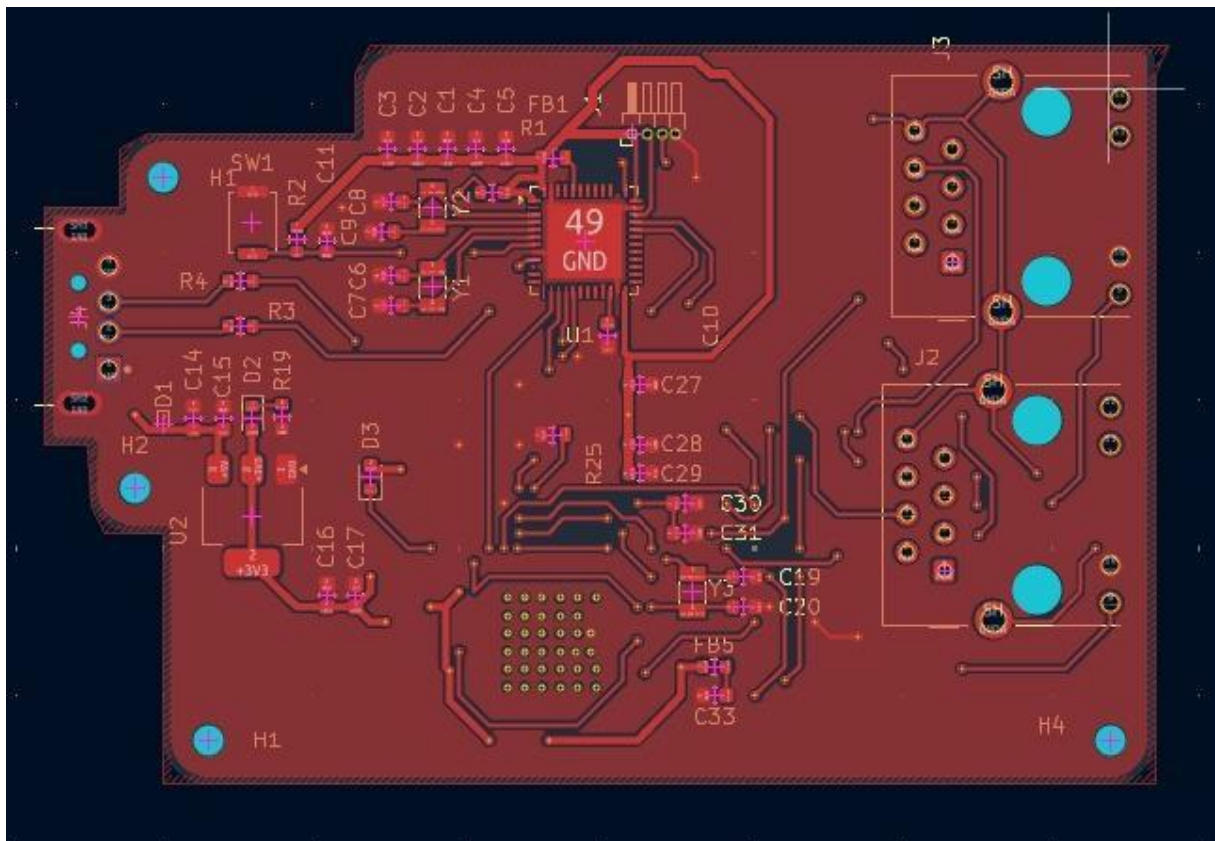


Figure 5 shows the top copper layer (F.Cu) of the PCB layout, including the routed tracks and connections between the components on the top layer.



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

Figure 6: Bottom Layer (B.Cu) of the PCB

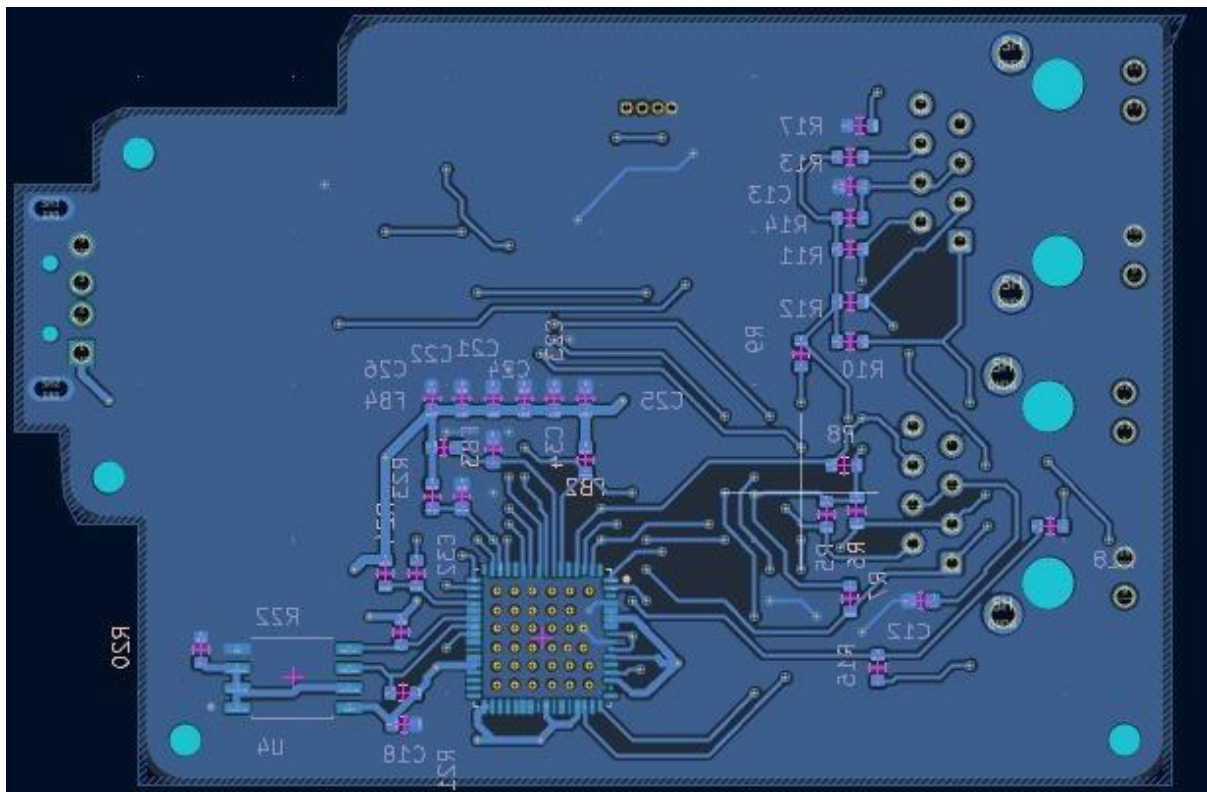


Figure 6 shows the bottom copper layer (B.Cu) of the PCB layout, including the routed tracks and electrical connections implemented on the bottom layer.



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

Figure 7: Top Assembly and Silkscreen View of the PCB

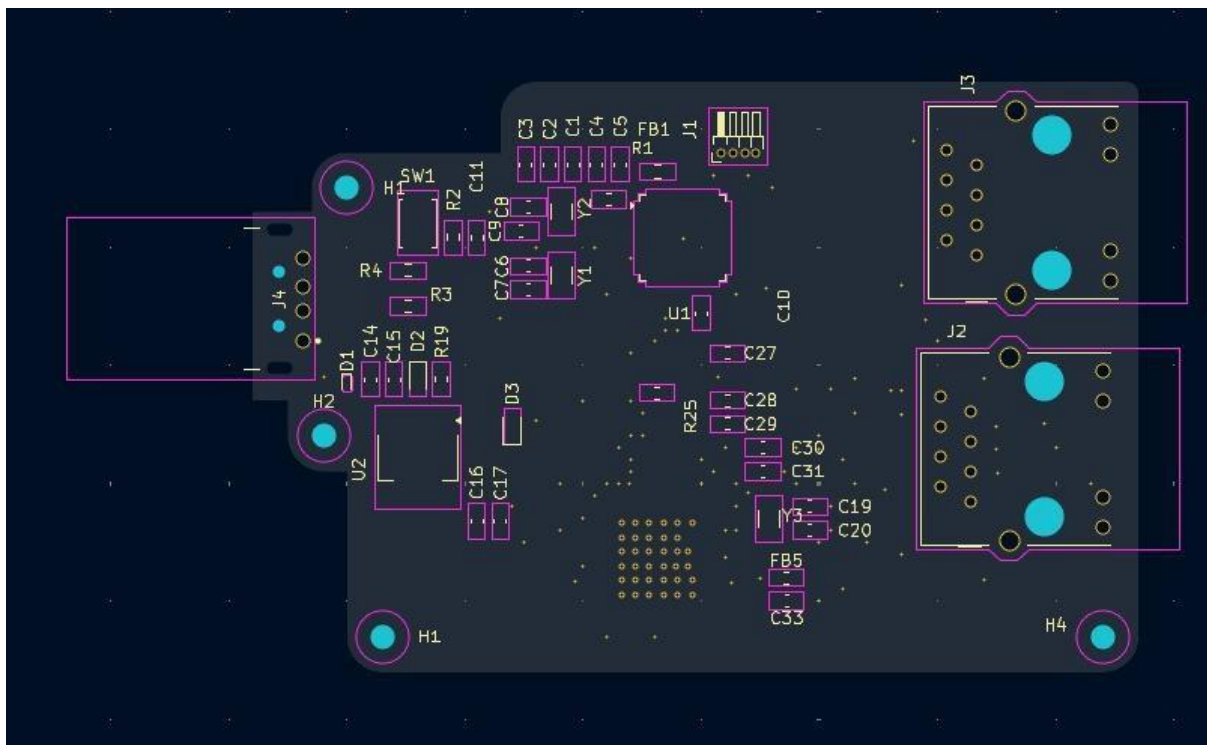


Figure 7 shows the top assembly and silkscreen view of the PCB. It displays the component placement and silkscreen markings on the top side of the board, helping to identify component locations and reference information during PCB assembly.



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

Figure 9: 3D Top View of the USB to EtherCAT Adapter Board

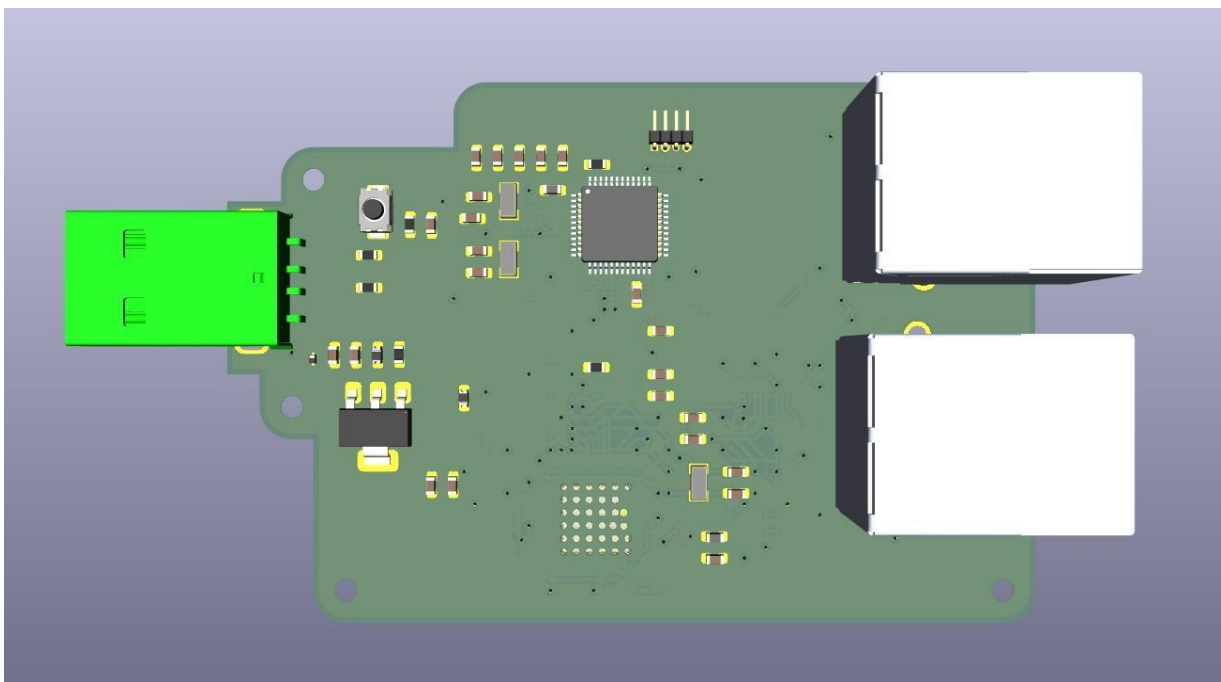


Figure 9 shows the three-dimensional top view of the completed USB to EtherCAT Adapter Board. It provides a visual representation of the PCB, component placement, board outline, connectors, and overall physical arrangement of the assembled design.



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

Figure 9: 3D Top View of the USB to EtherCAT Adapter Board

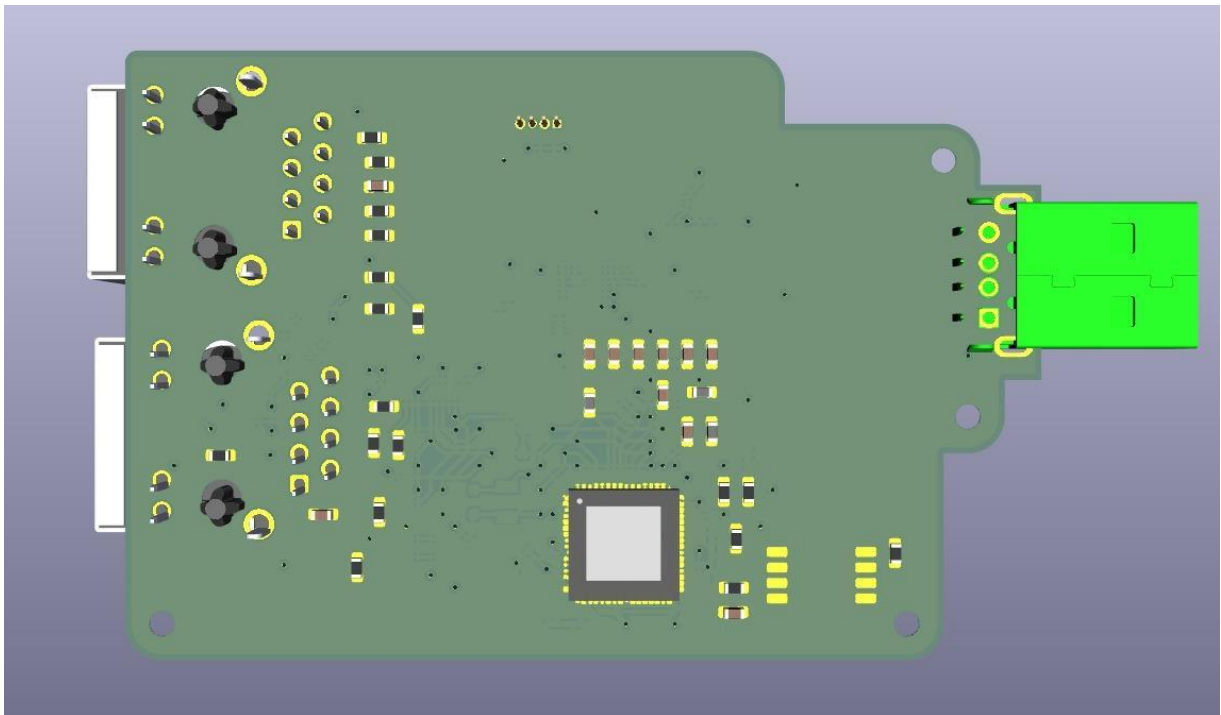


Figure 9 shows the three-dimensional top view of the completed USB to EtherCAT Adapter Board. It provides a visual representation of the PCB, component placement, board outline, connectors, and overall physical arrangement of the assembled design.



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

Figure 10: Side View of the USB to EtherCAT Adapter Board

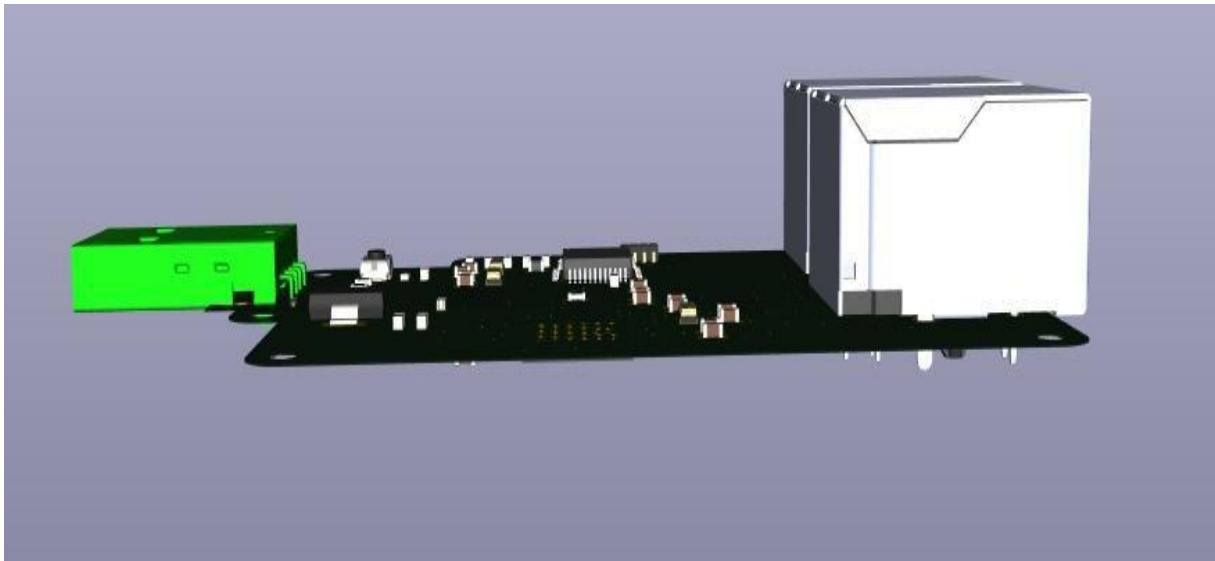


Figure 10 shows the side view of the USB to EtherCAT Adapter Board. It provides a view of the PCB thickness, component height, connectors, and overall physical arrangement of the board.



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

4.8 Gerber and Drill File Generation

After completing and verifying the PCB layout, the required manufacturing files were generated using KiCad PCB Editor.

Gerber files were generated for the required PCB fabrication layers, including copper, solder mask, silkscreen, and board outline layers. The required drill files were also generated for the holes in the PCB.

The generated Gerber and drill files were collected into a ZIP file for project submission and manufacturing purposes.

The generated files include:

- Front Copper (F.Cu)
- Back Copper (B.Cu)
- Front Solder Mask (F.Mask)
- Back Solder Mask (B.Mask)
- Front Silkscreen (F.Silkscreen)
- Back Silkscreen (B.Silkscreen)
- Board Outline (Edge.Cuts)
- Drill files

4.9 Bill of Materials (BOM)

A Bill of Materials (BOM) was generated from the project schematic using KiCad. The BOM provides a list of the components used in the PCB design and helps organize the component information required for procurement and assembly.

The BOM contains information such as:

- Reference designators
- Component values
- Quantities
- Component/package information, where available



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

5. SOCIAL / INDUSTRY RELEVANCE

The USB to EtherCAT Adapter Board project is relevant to the electronics, embedded systems, and industrial automation domains. EtherCAT is used in industrial communication systems where reliable and efficient communication between devices is important.

The project provides practical exposure to PCB development for communication-based electronic systems. The design process covers schematic development, PCB layout, routing, design verification, and preparation of manufacturing files.

The knowledge and skills gained through this project can be applied in areas such as:

- Industrial automation
- Embedded systems
- Communication interfaces
- Electronic hardware development
- PCB design and manufacturing

The project also provides practical experience in following a complete PCB design workflow, from schematic development to preparation of manufacturing files.



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

6. LEARNING & REFLECTION

Through this project, I gained practical knowledge of the complete PCB design process using KiCad. The project helped me understand how a circuit schematic can be developed into a complete PCB layout and prepared for manufacturing.

The major learning outcomes from the project include:

- Understanding and creating electronic schematics.
- Selecting and assigning appropriate component footprints.
- Designing PCB layouts using KiCad.
- Placing components and routing PCB tracks.
- Understanding PCB copper and other design layers.
- Performing Design Rule Checks (DRC).
- Generating Gerber and drill files.
- Generating a Bill of Materials (BOM).
- Using the KiCad 3D Viewer to inspect the physical PCB design.
- Organizing PCB project files and documentation for submission.

Project Reflection

Working on the USB to EtherCAT Adapter Board gave me hands-on experience with the complete PCB design workflow. I learned that proper component placement, routing, layer management, and design verification are important for developing a reliable PCB design.

The project also helped me improve my ability to use KiCad and understand the different stages involved in taking a PCB design from schematic development to manufacturing-file generation.



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

7. FUTURE SCOPE & CONCLUSION

7.1 Conclusion

The **USB to EtherCAT Adapter Board** project was successfully designed using KiCad. The project covered the major stages of PCB development, including schematic design, component placement, PCB layout, routing, design verification, Gerber and drill file generation, BOM preparation, and 3D visualization.

The completed PCB design was verified using the KiCad Design Rule Checker and achieved **zero DRC errors**. The required project and manufacturing files were also generated as part of the project deliverables.

This project provided valuable practical experience in PCB design and improved my understanding of the complete workflow involved in developing an electronic hardware design.

7.2 Future Scope

The project can be further developed through the following activities:

- Manufacturing the designed PCB.
- Assembling the components on the fabricated PCB.
- Performing electrical and functional testing.
- Testing USB communication.
- Testing EtherCAT communication.
- Performing hardware debugging and validation.
- Improving the PCB design based on practical testing results.
- Developing future revisions of the board based on test results.



*Innovation & Entrepreneurship Hub for Educated Rural Youth
(SURE Trust – IERY)*

SL. NO	ASSIGNMENTS	LINKS
1	Op-amp	https://drive.google.com/file/d/1oDjm_8cpMw7iBUZOvc7Owla40jjgsCoL/view?usp=drive_link
2	STM32F	https://drive.google.com/file/d/1MfCknawqEQ9yVSVy1CEactdGCUDVgw8/view?usp=drive_link
3	LDR	https://drive.google.com/file/d/1x0iX1iPluL66iLpnUW6C8X_I_gYd1XYE/view?usp=drive_link
4	Microcontroller	https://drive.google.com/file/d/1-t6eaG3zZTv1hfouPaikjAo-LeEQTBBL/view?usp=drive_link
5	Power Switch	https://drive.google.com/file/d/1tDh-wKkMK60Tt8gWx8Xe1J47wK9iahAx/view?usp=drive_link
6	LED Interfacing	https://drive.google.com/file/d/1GdWU9xmugZrkboqtfvWyoTawuYZrfZ1j/view?usp=drive_link
7	USB to EtherCAT Adapter board	https://drive.google.com/file/d/1KzfwmiAz-DatuJWne2Z5dChIWUAc_vUS/view?usp=drive_link